

Muhammad Ali

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EDUCATION

Merrimack College *Biochemistry (BS, Honors)*

North Andover, MA | Expected Graduation: May 2027

Relevant Courses: Cellular Biochemistry, Genetics, Organic Chemistry I & II (Honors), Analytical Chemistry, Inorganic Chemistry, Biophysical Chemistry, Directed Research, Physics II, Problem Solving with Java.

Community College of Baltimore County *Biology (AS, Honors) (Magna Cum Lauda)*

Baltimore, MD | August 2023 - May 2025

Relevant Courses: Biology I: Molecules & Cells, Evolution & Ecology, Anatomy & Physiology I, Microbiology, General Chemistry I & II, Introduction to Organic & Biochemistry, Physics I, Calculus I & II, Pre-Calculus II, Statistics (Honors).

Certificates Global Studies Certificate | Psychology Certificate

AWARDS

Dean's List X6 (Fall 2023, Spring 2024, Fall 2024, Spring 2025, Fall 2025, Spring 2026), Merrimack Honors Scholarship; Outstanding Future Warrior Scholarship; Phi Theta Kappa Scholarship; Presidential Honors Scholarship; Alex & Coppinger Scholarship; AJ Guckert Memorial Scholarship; L. Isenock Phi Theta Kappa Scholarship; Top 4% Student at Community College of Baltimore County.

RESEARCH EXPERIENCE

Keck Graduate Institute (KGI SURE), *Summer Research Intern*

Claremont, CA | 2026 Summer

- Selected for the highly competitive Summer Undergraduate Research Experience (SURE) Program in Drug Discovery and Business Development. Conducted interdisciplinary research evaluating therapeutic candidates and drug development
- Collaborated with faculty mentors and industry professionals to assess translational opportunities, commercialization pathways, and healthcare innovation. Analyzed scientific, clinical, and market considerations influencing pharmaceutical and biotechnology product development.

Fralin Biomedical Research Center (CardioSURF), *Undergraduate Research Intern, Virginia Tech*

Roanoke, VA | 2025 Summer

- Investigated the role of BMP3 signaling in pulmonary arterial hypertension using rat, mouse, pig, and human vascular cell models.
- Performed mammalian cell culture, RNA extraction, qPCR, ELISA, Western blotting, immunofluorescence microscopy, and cryosectioning.
- Analyzed gene and protein expression patterns associated with vascular remodeling and pulmonary hypertension progression.
- Collaborated with graduate students, postdoctoral fellows, and principal investigators in experimental design and data interpretation.

Merrimack College (SIRF Program), *Undergraduate Research Assistant*

North Andover, MA | August 2025 - Present

- Conduct synthetic medicinal chemistry research developing greener Mitsunobu methodologies for the synthesis of methyl isotopologues and deuterium-labeled therapeutics under Prof. Manikandan Santhanaraman.
- Synthesized tetrabenazine and deutetabenazine analogs using mechanochemical Mitsunobu reactions with bio-renewable solvents including γ -valerolactone (GVL) and 2-methyltetrahydrofuran (2-MeTHF).
- Focused on developing low-cost therapeutic analogs for Huntington's disease treatment. Investigated catechol, resorcinol, and methoxyphenol substrates, achieving product yields up to 92% under optimized conditions.

Boston University, *Undergraduate Research Assistant*

Boston, MA | December 2025 – Present

- Conduct qualitative coding and thematic analysis of participant responses for organizational psychology research under Prof. Richard A. Currie.
- Analyze survey datasets, identify recurring themes, and assist in data organization and interpretation. Contribute to research examining workplace behavior, employee experiences, and boundary expansion.

PUBLICATIONS & MANUSCRIPTS

A Mechanochemical Mitsunobu Strategy for the Greener Synthesis of Methyl Isotopologues (Muhammad Ali., Manikandan S.)

Expected 2026

Abstract Published and Poster presented at Northeast Student Chemistry Research (Simmons University, 2026), and Merrimack Annual Research & Creative Achievement Conference (2026).

Regulation of Bone Morphogenetic Protein 3 in Pulmonary Arterial Hypertension (Muhammad Ali., Aymen Halouani., Yassine Sassi)

August 2025.

Abstract Published and Poster Presented at American Heart Association Basic Cardiovascular Sciences (BCVS) Scientific Sessions (2025), Virginia Tech Research Symposium, (2025), Fralin Biomedical Research Institute Research Symposium (2025)

Cryptocurrency and Bitcoin: Technical Foundations, Historical Emergence, and Moral Significance (Muhammad Ali., Advised by Cronsted); Published Manuscript, Merrimack College (2026)

October 2025

Investigating the Chesapeake Bay: A Comprehensive Research on Environmental and Human Health Impacts (Muhammad Ali., Advised by Christina C.); Published in CCBC Journal Traverse

October 2024

TEACHING EXPERIENCE

Peer Tutor- Community College of Baltimore County

Baltimore, MD | May 2024 – May 2025

- Level II Tutor- Completed 500+ hours of tutoring across STEM disciplines through the Academic Learning Center Provided individual and group instruction in Calculus I & II, Statistics, Physics I, General Chemistry, Biology, Microbiology, and Genetics. Assisted students with problem-solving strategies, exam preparation, laboratory concepts, and study skills.

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PROJECTS

Velyra Health, Founder (<https://velyra-health.lovable.app>)

Boston, MA | November 2025 - Present

- Developing a zero-contact patient monitoring platform using AI, computer vision, and sensor-based technologies for continuous vital sign monitoring.
- Designed remote patient monitoring workflows for heart rate, respiratory rate, oxygen saturation, temperature, sleep, and movement tracking
- Presented and pitched the venture at the Merrimack College Pitch Competition.

TIDE (Tidal Health), Project (<https://tidal-health-insight.lovable.app>)

Cambridge, MA | Oct 2025 - Present

- Developed a healthcare innovation focused on congenital heart disease screening and monitoring.
- Performed market research, clinical workflow design, and healthcare implementation analysis

Harmona Health, Co-founder

Cambridge, MA | October 2025 - Present

- Developed an AI-enabled healthcare platform integrating EEG, ECG, and large language models to support personalized mental health assessment.
- Collaborated with interdisciplinary teams during MIT Hacking Medicine GrandHack to design predictive healthcare solutions. Presented the project to clinicians, healthcare innovators, and industry mentors.

PressSure, Team Member

Cambridge, MA | September 2025

- Developed a blood pressure monitoring and healthcare accessibility solution. Conducted clinical need assessment, healthcare market analysis, and solution development.

PAH Insight (Pulmonary Arterial Hypertension), Project (<https://pah-insight.lovable.app>)

Baltimore, MD | August 2025

- Developed an educational and research-focused platform on Pulmonary Arterial Hypertension (PAH).
- Integrated disease information, research findings, and clinical insights into an interactive web application.

Mystery Word Game, Course Project

Remote | December 2025

- Designed and implemented a console-based puzzle game in Java using object-oriented programming principles.
- Implemented CSV-based word bank integration, recursion, file I/O, score tracking, and JUnit testing.

SERVICE & LEADERSHIP

Community College of Baltimore County: Student Government Association (SGA) - President, Student Honors Council - Vice President, International Student Association- President, Student Ambassador, Phi Theta Kappa Society, Fresh Faces Initiative 2024-25
Merrimack College: Resident Advisor (RA), Global Ambassador, Security Building, Phi Theta Kappa Honor Society 2025-2027
Volunteering: FIFA World Cup, Red Cross, Shaukat Khanum.

CLINICAL EXPERIENCE

Emergency Medical Technician-EMT Trainee, National EMS Institute

Thorndale, PA | 2026

- Completed EMT-B training in patient assessment, emergency response, trauma management, CPR/BLS, airway management, and prehospital emergency care.

Cardiology Shadowing – Dr. Hassan Kassam Ali, MD

Baltimore, MD | 2024

- Completed 150+ hours of physician shadowing in cardiology, observing patient evaluations, ECG interpretation, stress testing, echocardiography, cardiovascular diagnostics, and clinical decision-making.

FELLOWSHIPS & INTERNSHIPS

Summer Undergraduate Research Education Program (SURE), University of Georgia

Remote | Summer 2026

Berkeley Pharma Tech, Editorial Intern, University of California, Berkeley

Berkeley, CA | Spring 2026

Molecular Imaging Mini-Fellowship, Stanford University

Remote | Summer 2025

ThinkNeuro, Bibliometrics Research Intern

Remote | August – December 2025

Technology in Healthcare Program, Stanford University

Remote | August – December 2025

Undergraduate Plummer Scholar, Mayo Clinic

Remote | 2025-2026

Rising Docs Scholar, New York University

Remote | 2025-2026

SKILLS & INTERESTS

Laboratory Techniques: Mammalian Cell Culture, qPCR, Western Blotting, ELISA, Immunofluorescence Microscopy, RNA Extraction, Immunostaining, Micro pipetting, Cryosectioning, HPLC, NMR Spectroscopy, Thin Layer Chromatography (TLC), Preparative TLC, Flash Chromatography (CombiFlash), UV-Vis Spectroscopy, Fluorescence Spectroscopy, Gas Chromatography (GC), Atomic Absorption Spectroscopy (AAS), Sample Preparation, Molecular Biology Techniques, Analytical Chemistry Techniques, Synthetic Organic Chemistry, Medicinal Chemistry, Molecular Imaging Exposure (PET, MRI, Optical Imaging)

Research Skills: Experimental Design, Scientific Writing, Literature Review, Manuscript Preparation, Poster Development, Scientific Presentations, Data Collection, Data Analysis, Statistical Analysis, Qualitative Coding, Bibliometric Analysis, Research Communication, Academic Peer Review

Languages: Python, R, Java,

Programming & Data Science: Microsoft Excel, Google Sheets, Data Visualization, Machine Learning Fundamentals, Predictive Modeling, Feature Engineering, Data Cleaning, Computational Analysis